

Evaluating Fairness in ICU Mortality Prediction with MIMIC-IV (Midterm Report)

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(Dated: April 3, 2025)

I. Dataset Description

This study utilizes the Medical Information Mart for Intensive Care (MIMIC-IV), a publicly available database containing de-identified health-related data from over 50,000 ICU admissions at the Beth Israel Deaconess Medical Center between 2008 and 2019. MIMIC-IV includes detailed information on demographics, vital signs, laboratory measurements, treatment interventions, and clinical outcomes. Its extensive coverage and diversity allow for developing predictive models and examining their performance across demographic groups, enabling analysis of potential biases and fairness. The dataset's granularity and diversity provide an opportunity to explore how AI models perform across demographic groups and assess disparities in predictive outcomes. [6] [7] [1]

Research Questions

This project aims to address the following **primary research question**:

How can AI models be developed and evaluated to ensure fairness in ICU mortality prediction concerning MIMIC-IV sensitive attributes?

The following **supporting sub-questions** will guide this research:

1. What sensitive attributes in the MIMIC-IV dataset contribute to biased outcomes in ICU mortality prediction?
2. How do Logistic Regression, XGBoost, and Multilayer Perceptron (MLP) models differ in predictive performance and fairness across demographic groups?
3. Which fairness metrics are most effective for evaluating biases in ICU mortality prediction models?
4. What bias mitigation strategies can improve model fairness without compromising predictive accuracy?

Objectives

This project aims to develop predictive models using Logistic Regression(LR), XGBoost, and Multilayer Perceptron (MLP) to forecast ICU mortality. It focuses on identifying potential biases within these models, specifically examining sensitive attributes such as

race, gender, age, insurance type, ethnicity, language preference, marital status, and socioeconomic status. To evaluate model fairness, the project will utilize fairness metrics, including demographic parity, equal opportunity, equalized odds, disparate impact, and calibration across subgroups. Additionally, performance metrics like AUC(min), AUC(macro-avg), and AUC(minority) will assess disparities, with equalized odds difference measuring consistency in true and false positive rates.

The project also proposes bias mitigation strategies to improve model fairness while maintaining predictive accuracy. These strategies include reweighting (adjusting training sample weights), threshold modification (adjusting decision thresholds), and the Synthetic Minority Oversampling Technique (SMOTE) to address class imbalances. Ultimately, this research seeks to advance the development of equitable AI systems in healthcare, ensuring predictive models support unbiased decision-making, optimize resource allocation, and reduce the cognitive burden on healthcare professionals. [5] [4] [2]

Data Availability

The data used in this study is available from the MIMIC-IV database (version 3.1), which can be accessed at <https://physionet.org/content/mimiciv/3.1/>. Access to the dataset requires credentialed approval through the PhysioNet platform after completing the required data use agreements and training on human subjects research.

II. Methodology

Data Preprocessing and Feature Engineering

The study employs a comprehensive data preprocessing pipeline implemented in Python using the Pandas library. The analysis incorporates six primary tables from the MIMIC-IV database, each contributing distinct clinical and demographic information:

1. Demographics of the patient (core/patients.csv)
2. Hospital admission records (core/admissions.csv)
3. Intensive Care Unit stay information (icu/icustays.csv)
4. Laboratory measurements (hosp/labevents.csv)
5. Vital signs and additional charted data (icu/chartevents.csv)
6. Patient output measurements (icu/outputevents.csv)

1. Feature Selection and Engineering

The study examines several sensitive attributes to assess potential algorithmic bias: - Demographic characteristics (age, gender) - Socioeconomic indicators (insurance type) - Cultural factors (language preference, ethnicity) - Social determinants (marital status)

All categorical variables undergo one-hot encoding to facilitate model compatibility and interpretation.

For clinical measurements, we implement a systematic aggregation approach:

1. Laboratory values: Statistical features including mean, standard deviation, minimum, and maximum values are computed to capture the distribution and range of measurements
2. Vital signs: Similar statistical aggregations (mean, standard deviation, minimum, maximum) are performed to represent physiological trends
3. Output events: Cumulative sum and mean values are calculated to quantify patient output patterns

2. Target Variable Construction

The target variable for mortality prediction is determined by analyzing the relationship between patient death time and ICU discharge, specifically identifying cases where death occurred during the ICU stay. The binary classification outcome serves as the dependent variable for our supervised learning models.

3. Data Integration and Preprocessing Pipeline

The main preprocessing pipeline includes several critical steps:

1. Feature integration: Merging of multiple data sources while maintaining data integrity
2. Missing value imputation: Implementation of appropriate strategies for handling incomplete data
3. Feature scaling: Standardization of numerical features to ensure comparable scales
4. Dataset partitioning: Stratified splitting of data into training and testing sets while preserving class distributions

4. Future Feature Engineering Considerations

The methodology allows for future refinement through several potential measures: 1) Incorporation of additional clinical parameters 2) Refinement of aggregation methodologies 3) Integration of temporal feature extraction 4) Development of feature interaction terms

A. Technical Implementation Note

The preprocessing pipeline is implemented using Python's scientific computing stack, primarily utilizing the Pandas library for data manipulation and scikit-learn for preprocessing operations. This ensures reproducibility and maintains computational efficiency when handling large-scale healthcare data.

III. Implementation

A. Data Preprocessing and Feature Engineering

In the initial phase of the project, we focused on loading and preparing data from the MIMIC-IV dataset to establish a robust foundation for predictive models addressing ICU mortality. The MIMIC-IV dataset provides comprehensive patient demographics, clinical measurements, and ICU stay details.

1. Loading the Data

We utilized key tables from MIMIC-IV including patients.csv, admissions.csv, icustays.csv, diagnoses.csv, chartevents.csv, labevents.csv, d_items.csv, and d_labitems.csv. These tables provided essential data for analyzing sensitive features, group fairness, and gender modeling, and for defining our target variable (mortality).

File	What it gives you
patients.csv.gz	Gender, anchor age, subject ID
admissions.csv.gz	Admission + discharge time, insurance, ethnicity, language, hospital outcome (mortality)
icustays.csv.gz	ICU stay info: duration, ICU type, stay_id (useful for matching with charted data)

FIG. 1. Tables contain sensitive features, group fairness and gender modeling

The above files 1 are useful for sensitive features, group fairness and gender modeling.

File	Use
diagnoses_icd.csv.gz	Diagnosis codes — useful to create comorbidity indices or target labels
transfers.csv.gz	Bed transfers (can indicate care complexity or multiple ICU stays)

FIG. 2. Tables help define the target variable

The above tables2 help define the target variable (Y) like mortality, which is crucial to our study.

File	Use
chartevents.csv.gz	Vital signs, ventilator settings (heart rate, BP, GCS, etc.)
labevents.csv.gz	Lab results like WBC, sodium, BUN, creatinine, etc.

FIG. 3. Lab and Experiment Data

The above two tables3 contain lab and experiment data

while patients are in ICU. Due to the size of chartevents and labevents, chunk loading was necessary.

2. Data Filtering

To focus analysis on relevant clinical measurements, filtering was performed:

- **Vital signs:** Filtered using itemids from the d_items table under the category 'Routine Vital Signs' in chartevents.
- **Lab measurements:** Managed through a two-step process in labevents. Initially, entries without numeric values were dropped. Subsequently, we retained only the top five most frequently occurring lab itemids to optimize analytical relevance.

Filtered data were consolidated, and timestamps (charttime) were standardized to datetime format.

3. Handling Missing Data

Missing values in the dataset were systematically identified and addressed.

- **Missing values contains information:** For the attribute dod (date of death), missing values indicate that the patient's date of death was not recorded, suggesting the patient was not documented as deceased during the ICU stay. Given the importance of this information as an indicator, these missing values were intentionally retained.
- **Essential timestamps:** Rows missing outtime in icustays were removed, as these were critical for determining ICU mortality.
- **Non-imputable data:** Significant missing data in the valuenum column of labevents and chartevents led to removal rather than imputation. Initial attempts to impute these missing values with mean and median methods resulted in computational difficulties due to the dataset's substantial size. Consequently, rows with missing or non-numeric valuenum entries were excluded. Following this removal, substantial data remained for robust analysis (over 21 million lab entries and 27 million chart entries).
- **Categorical attributes:** Columns like insurance, language, and marital.status had missing values imputed as 'UNKNOWN'. Given extensive missing data in these categories, our analysis prioritized gender, age, and race as sensitive attributes.

4. Data Encoding

Key categorical data transformations were performed to optimize computational efficiency:

- **Gender:** Converted to binary encoding. Gender data was converted into binary format to enhance computational efficiency and simplify categorical handling in subsequent machine learning analyses.
- **Race:** Standardized using the map_race() function, merging similar categories⁴. This process effectively addressed inconsistencies and variations in how racial and ethnic data were originally documented, such as combining the categories Hispanic and Latino.

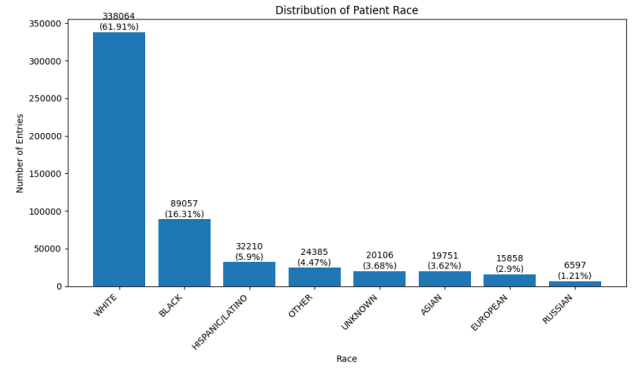


FIG. 4. Race Consolidation

- **Age grouping:** anchor_age was segmented into age groups ("18-29", "30-49", "50-64", "65-79", "80-91")⁵. Grouping ages in this manner enhances model interpretability and efficiency by reducing dimensional complexity, which is particularly beneficial for analyses using logistic regression, gradient-boosted models (e.g., XGBoost), and deep learning frameworks. This approach facilitates the detection of age-related trends and simplifies model training and evaluation.

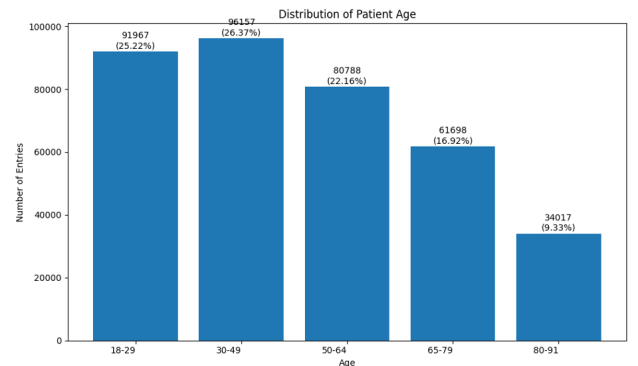


FIG. 5. Age Grouping

5. Merging Tables

We merged datasets to create a comprehensive dataframe comprising 94,444 entries with 34 columns, including sensitive attributes (gender, anchor_age, race_grouped). ICU mortality was calculated using date of death (dod) and ICU discharge time (outtime). Chartevent and labevent measurements taken during ICU stays were aligned using intime and outtime for accurate temporal analysis.

Vital signs selected based on predictive relevance and dataset frequency included:

1. Heart Rate (HR, itemid 220045)
2. Respiratory Rate (itemid 220210)
3. Systolic Blood Pressure (itemid 220277)
4. Diastolic Blood Pressure (itemid 220179)
5. Temperature (itemid 220180)

Vital signs and lab results were aggregated (mean and standard deviation) and merged into the main dataframe. Missing values were flagged and replaced with placeholder -1, ensuring data completeness.

B. Model Training and Evaluation: Logistic Regression with SMOTE

In this phase of the project, we integrated several datasets to create a comprehensive dataframe with 94,444 rows and 34 columns, including three sensitive attributes gender, anchor_age, race_grouped. To compute the target variable, ICU mortality, we identified the dod (date of death) and outtime (ICU discharge time) after joining the relevant tables.

Additionally we identify the chartevent and labevent that only measure during the patients stay in the icu by comparing intime outtime and chartime.

Also selecting the top 5 chart events for vital signs, which were identified based on their relevance to ICU mortality prediction and their frequency in the dataset. The selected chart events are as follows:

- 220045 - Heart Rate (HR): A crucial vital sign that measures the patient's heart rate, typically recorded in beats per minute (bpm).
- 220210 - Respiratory Rate: This event records the patient's respiratory rate, critical for assessing the patient's respiratory function.
- 220277 - Systolic Blood Pressure: This event tracks the patient's systolic blood pressure, a key indicator of cardiovascular health and circulatory function.

- 220179 - Diastolic Blood Pressure: This event measures diastolic blood pressure, which, together with systolic blood pressure, provides an overall picture of the patient's blood pressure status.
- 220180 - Temperature: This event records the patient's body temperature, an important measure for detecting infections or other health complications.

The chart event data for vital signs were filtered, aggregated, and merged into the primary dataset. Mean and standard deviation calculations were performed for vital signs and lab test variables, including heart rate, respiratory rate, blood pressure, and temperature. These aggregated values were integrated into the main dataframe.

Following a left merge, missing values were flagged and replaced with a placeholder (-1), ensuring comprehensive feature coverage. The resulting dataset now contains enriched patient demographics, ICU stay details, vital signs, and lab tests, making it suitable for mortality prediction models by capturing essential clinical indicators.

We trained a Logistic Regression model to predict ICU mortality, applying SMOTE (Synthetic Minority Over-sampling Technique) to address class imbalance and standardizing continuous features. After dividing the data into training and test subsets, the balanced dataset was used for model training.

Model performance was assessed through precision, recall, F1-score, and AUC metrics.⁶ Adjusting the classification threshold to 0.4 showed a modest drop in accuracy but improved precision for the majority class and recall for the minority class.⁷ Despite these improvements, precision for the minority class remained low, and AUC slightly decreased, highlighting the trade-off between accuracy and sensitivity.

Classification Report for df_data:					
	precision	recall	f1-score	support	
0	0.96	0.86	0.91	17339	
1	0.27	0.61	0.38	1550	
accuracy			0.84	18889	
macro avg	0.62	0.73	0.64	18889	
weighted avg	0.90	0.84	0.86	18889	
AUC for df_data: 0.8124580983760271					

FIG. 6. Classification Report

In summary, SMOTE effectively balanced the dataset and enhanced recall for the minority class but did not fully address the persistently low precision issue.

IV. Related Works

MIMIC-IV Dataset Usage

The MIMIC-IV dataset has demonstrated significant utility across various clinical research applications.

Threshold Adjustment Classification Report for df_data:					
	precision	recall	f1-score	support	
0	0.97	0.79	0.87	17339	
1	0.23	0.70	0.35	1550	
accuracy			0.78	18889	
macro avg	0.60	0.75	0.61	18889	
weighted avg	0.91	0.78	0.83	18889	
AUC (threshold adjusted) for df_data: 0.7456325568502109					

FIG. 7. Threshold Adjustment Classification Report

Zhang et al. [13] developed a mortality predictive model for ICU patients with primary pulmonary hypertension (PPH), incorporating multiple physiological parameters including age, breathing rate, blood characteristics, glucose levels, and SAPS II severity scores. Their model demonstrated robust discriminative capability between high- and low-risk patients, with validated clinical utility.

In the domain of pharmacovigilance, Wei et al. [11] conducted research on medication-related information extraction, focusing on identifying drug names and their associated adverse effects from medical records. Vistisen et al. [10] utilized the dataset to investigate the prevalence and temporal distribution of extrasystoles in septic ICU patients, specifically examining their potential as predictors of fluid responsiveness. Additionally, Tasnim and Mamun’s research [9] evaluated the impact of feature selection methodologies on ICU mortality prediction models, demonstrating that Principal Component Analysis (PCA) enhanced predictive accuracy across multiple machine learning classifiers including Logistic Regression, Decision Tree, K-Nearest Neighbors, and Support Vector Machine algorithms.

Related Methods to Research QuestionsI

Model Performance Comparison Studies

Wei et al. [11] implemented a comprehensive system comprising two main components: named entity recognition (NER) and relation classification (RC). Their methodology involved comparing deep learning-based approaches, particularly BI-LSTM-CRF, against traditional machine learning methods such as conditional random fields for NER and support vector machines for RC. Similarly, Tasnim and Mamun [9] conducted comparative analyses of feature selection techniques, specifically employing PCA across various classification models including Logistic Regression, Decision Tree, K-Nearest Neighbors, and Support Vector Machine.

Bias and Fairness in Clinical Prediction Models

Recent research has increasingly focused on addressing algorithmic bias in clinical prediction models. Yang et al. [12] introduced an adversarial training framework designed to mitigate biases while maintaining high clinical performance, demonstrating generalizability across various outcomes, models, and fairness definitions through validation in multiple hospital cohorts. Sufian et al. [8] addressed algorithmic bias in cardiovascular risk prediction models by integrating fairness-aware algorithms, SCIR models, and interpretability frameworks. Their research demonstrated that bias mitigation techniques successfully improved fairness metrics while maintaining strong predictive accuracy. Furthermore, Gu et al. [3] investigated demographic biases in COVID-19 mortality prediction models, evaluating transfer learning’s effectiveness in improving model fairness across diverse racial and ethnic groups, with particular success in enhancing predictive performance for Non-Hispanic Black, Hispanic/Latino, and Asian populations.

Novelty Analysis

Building upon existing literature in clinical prediction models and fairness-aware machine learning, this research addresses critical gaps in the comprehensive analysis of fairness-performance trade-offs in ICU mortality prediction. While prior work by Yang et al. [12] and Gu et al. [3] explored bias mitigation through adversarial training and transfer learning approaches, these studies primarily focused on racial bias or theoretical frameworks, without utilizing the MIMIC dataset’s rich clinical data.

This research advances the field through three key contributions. First, it provides a novel comparative analysis of fairness-performance trade-offs across Logistic Regression, XGBoost, and Multilayer Perceptron architectures in ICU mortality prediction. Second, it implements a comprehensive evaluation framework incorporating multiple fairness metrics, surpassing the typical one or two metrics used in existing studies. Third, it introduces an innovative end-to-end framework that integrates sensitive attribute analysis, model comparison, fairness evaluation, and bias mitigation strategies—components often addressed separately in current literature.

Through this integrated approach, the research not only extends existing work in fairness-aware clinical prediction models but also addresses fundamental gaps in methodology and practical implementation, contributing to the development of more equitable clinical decision support systems.

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